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Fibonacci number f_n,

$A_1 r_1^n + A_2 r_2^n$

$f_0 = f_1 = 1$

$f_n = f_{n-1} + f_{n-2}$

Assume, $a_n = Ar^n$

$Ar^n = Ar^{n-1} + Ar^{n-2}$

$\Rightarrow r^2 - r - 1 = 0$

$\Rightarrow r = \frac{1 \pm \sqrt{5}}{2}$

General soln : $A_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + A_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$.

$A_1 + A_2 = 0$ $\quad f_0 = 0, \; f_1 = 1$

$$A_1\left(\frac{1+\sqrt{5}}{2}\right) + A_2\left(\frac{1-\sqrt{5}}{2}\right) = 1$$

$$\Rightarrow \quad A_1(\sqrt{5}) = 1$$

$$A_1 = \frac{1}{\sqrt{5}}, \quad A_2 = \frac{-1}{\sqrt{5}}$$

$f_0 = f_1 = 1$

$$A_1 = \frac{\sqrt{5}+1}{2\sqrt{5}}, \quad A_2 = \left(\frac{\sqrt{5}-1}{2\sqrt{5}}\right)$$

General soln $A_1\left(\frac{1+\sqrt{5}}{2}\right)^n + A_2\left(\frac{1-\sqrt{5}}{2}\right)^n$

$a_n = 2a_{n-1} - a_{n-2}$

$a_1 = 2, \; a_2 = 3$

Assume $a_n = Ar^n$

$Ar^n = 2Ar^{n-1} - Ar^{n-2}$

$\Rightarrow \; r^2 - 2r + 1 = 0$

$\Rightarrow \; r = 1, 1$ roots $= 1, 1$

$A(1)^n$ whatever A, will not satisfy both base cases (problem: cannot fit boundary conditions)

General soln: $An r^n$ ✗

$a_n = (An+B)r^n$

$a_n = 2a_{n-1} - a_{n-2}$

$\Rightarrow (An+B)r^n = 2(A(n-1)+B)r^{n-1} - (A(n-2)+B)r^{n-2}$

$\Rightarrow (An+B)r^2 - 2(A(n-1)+B)r + (A(n-2)+B) = 0$

$\Rightarrow B(r-1)^2 + An(r-1)^2 + 2A(r-1) = 0$ $\quad B = 1$

$An r^2 - 2A(n-1)r + A(n-2) = 0$

$(r-1)[(An+B)(r-1) + 2A] = 0$

$n(Ar^2 - 2Ar + A) + 2Ar - 2A = 0$

~~x~~ combination of roots turn out to be gen. soln.

Now, checking, whether $(An+B)1^n$ also, gen. soln (under $r=1$).

gen. soln $= (An+B)$

$A=1, B=1$

$\boxed{a_n = n+1}$

n

$$D_n = \begin{vmatrix} 2 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & \cdots \\ 0 & 1 & 2 & 1 & \\ 0 & 0 & 1 & 2 & \\ & \vdots & & & \end{vmatrix}_{n \times n}$$

$D_n = 2 \times \det(D_{n-1}) - 1 \cdot \det \begin{bmatrix} 1 & 1 & 0 & \cdots \\ 0 & 2 & 1 & 0 \cdots \\ 0 & 1 & 2 & 1 \cdots \\ 0 & 0 & 1 & 2 \cdots \\ \vdots & & & \end{bmatrix}_{n-1}$

$D_1 = 2$

$D_2 = 3$

$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$\{ \det A_{n \times n} = \det A^T_{n \times n} \}$

$= 2 \times (D_{n-1}) - 1 \cdot (D_{n-2})$

$\boxed{D_n = 2D_{n-1} - D_{n-2}}$

$D_1 = 2$

$D_2 = 3$

$D_n = n+1$

Quadratic
homogeneous
real, distinct
real, equal
complex

$$D_n = \det \begin{vmatrix} b & b & 0 & 0 & \cdots \\ b & b & b & 0 & \cdots \\ 0 & b & b & b & \cdots \\ 0 & 0 & b & b & \cdots \end{vmatrix}$$

} shifting 3 b's from 2nd row

$D_1 = b$

$D_2 = 0$

$|b|$

$\begin{vmatrix} b & b \\ b & b \end{vmatrix}$

$$= b D_{n-1} - b \begin{vmatrix} b & b & 0 & \cdots \\ b & b & b & \cdots \\ 0 & b & b & \cdots \\ 0 & b & b & \cdots \end{vmatrix}$$

$$= b D_{n-1} - b \cdot b D_{n-2}$$

$$D_n = (b D_{n-1}) - (b^2 D_{n-2})$$

Assume: Ar^n

$$r^2 - br + b^2 = 0$$

$$r = \frac{b \pm \sqrt{b^2 - 4b^2}}{2}$$

$$= b\left(\frac{1 \pm i\sqrt{3}}{2}\right)$$

2 conjugate roots.

General soln.

$$A r_1^n + A_2 r_2^n$$

& solve

A_1, A_2 - real values.

$z^n = r^n(\cos n\theta + i \sin n\theta)$.

$D_1 = b$

$D_2 = 0$

$r_1 = A_1(\cos\theta_1 + i\sin\theta_1)$

$r_2 = A_2(\cos\theta_2 + i\sin\theta_2)$.

- Linear Hom. recurrence

$a_n = c a_{n-1}$, $a_0 = B$.

$a_n = B c^n$.

Quadratic Hom. recurrence:

$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} = 0$.

(Similar to above) $a_n = Ar^n$

$c_0 a^2 + c_1 a + c_2 = 0$

Real, Dist. (r_1, r_2) — Gen. soln $A_1 r_1^n + A_2 r_2^n$

Complex — to make form simpler: $a+ib = r(\cos\theta + i\sin\theta)$

Real, equal — Gen. soln $(A_1 n + B) r^n$

guess, extend again checks

Hom. recurrence:

$c_0 a_n + c_1 a_{n-1} + \cdots + c_k a_{n-k} = 0.$

Assume, $a_n = Ar^n$

$a_0 = d_0$, $a_{k-1} = d_{k-1}$

$c_0 r^k + c_1 r^{k-1} + \cdots + c_k = 0.$

($a_n = c a_{n-1}$, adding lower terms, still soln same form)

k roots

General soln: $A_1 r_1^n + A_2 r_2^n + \cdots + A_k r_k^n.$

$r_1 = r_2$

$(A_1 n + B_1) r_1^n + A_2 r_2^n + \cdots A_k r_k^n$

$r_1 = r_2 = r_3$

$(A_1 n^2 + B_1 n + C_1) r_1^n + A_4 r_4^n + \cdots + A_k r_k^n.$

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* Non-homogeneous recurrence:

$a_0 = A$

$a_n = c a_{n-1} + f(n)$

when $c = 1$

$a_n = a_{n-1} + f(n)$

$= a_{n-2} + f(n-1) + f(n)$

$= a_0 + f(1) + f(2) + \cdots f(n)$

$= A + \sum_{i=1}^{n} f(i)$

Bubble Sort: $a_1 = 0$

$a_n = a_{n-1} + (n-1)$

$a_n = 0 + \sum_{i=1}^{n} (i-1) = \frac{n(n-1)}{2}$

Selection Sort:

↳ Number of cmps → same,

$\frac{n(n-1)}{2}$.

Merge Sort:

$$a_n = 2a_{n/2} + n$$ → later

$a_n = c a_{n-1} + f(n)$ $\quad a_0 = \lambda$.

$f(n) = d r^n$.

$\Rightarrow a_n = c a_{n-1} + d r^n$.

$= c[c a_{n-2} + d r^{n-1}] + d r^n$

$= c^2 a_{n-2} + (c) d r^{n-1} + d r^n$ → dominated by r^n.

particular soln: $a_n^{(p)} = A r^n$.

$A r^n = (cA r^{n-1}) + (d r^n)$ → from this assumed particular soln.

↓ dominated by $d r^n$

$\Rightarrow Ar = cA + dr$

$\Rightarrow \boxed{A = \frac{dr}{r-c}}$

$a_n^{(h)}$: $a_n = c a_{n-1}$ $\quad c^n = \lambda$.

$a_n = B \cdot a_n^{(h)} + a_n^{(p)}$

$= B c^n \lambda + \left(\frac{dr}{r-c}\right) r^n$.

$\lambda = a_0 = B \cdot \lambda + \frac{dr}{r-c}$

↳ lets us satisfying boundary condition.

↳ satisfy recurrence

gen. soln for $a_n = a_n^{(h)} + a_n^{(p)}$

$$\boxed{a_n = c^n \lambda + \left(\frac{dr}{r-c}\right) r^n}$$

Eg: $a_n = 3a_{n-1} + 5 \cdot 7^n$, $\quad a_0 = 1$.

$a^{(p)} = A \cdot 7^n = \frac{35}{4} \cdot 7^n$

$\Rightarrow A(7) = 3 \cdot A + 5 \cdot 7$

$A = \frac{35}{4}$

$a_n^{(h)}$, $a_n = 3a_{n-1}$, $a_0 = 1$

$a_n = 3^n$.

$a_n = a_n^{(h)} + a_n^{(p)}$

$= \lambda(3^n) + \frac{35}{4} \cdot 7^n$

$a_0 = 1 = \lambda + \frac{35}{4}$

$\Rightarrow \lambda = -\frac{31}{4}$

$a_n = -\frac{31}{4} \cdot 3^n + \frac{35}{4} \cdot 7^n$

$\frac{dv}{v-c} = \frac{5(3)}{3}$ $a_n = 3a_{n-1} + 5 \cdot 3^n$, $a_0 = 1$.

$A \cdot 3 = 3 + 5 \cdot 3$

$A = 6$

$3A = 8$

$A = \frac{8}{3}$

$A \cdot 3^n$.

$A \cdot 3^n = 3A \cdot 3^{n-1} + 5 \cdot 3^n$

$0 = 0 + 5 \cdot 3^n$

$a_n^{(p)} = A \cdot 3^n \rightarrow a_n^{(p)} = An \cdot 3^n$

$a_n^{(h)} = B \cdot 3^n$

$a_n = a_n^{(p)} + a_n^{(h)}$

$A \cdot 3^n = A \cdot 3^n + B \cdot 3^n$ $(A+B)^n$.

$= (A+B)3^n$

But $A \cdot n$ not even satisfying recurrence.

Base case still does not work.

$a_n^{(p)} \neq An3^n$

$a_n^{(h)} = B \cdot 3^n$ only to meet base condition

$An3^n + B3^n$

so, modification in $a_n^{(p)}$

$= An \cdot 3^n$

$(An+C)3^n$ going to give same as $An \cdot 3^n$

$A n \cdot 3^n = 3A(n-1)3^{n-1} + 5 \cdot 3^n$ ... (1)

$\Rightarrow 3An = 3A(n-1) + 5n(3)$

$\Rightarrow 3An = 3An - 3A + 5n(3)$

$\Rightarrow 3A = 5n$ $\quad A = 5$

$A = \frac{5n}{3}$

$5n\,3^n + B \cdot 3^n$

$a_0 = 1$ $\quad B = 1$

$a_n = 5n\,3^n + 3^n$

$a_n = 3^n(5n+1)$

$f(n) = d \cdot r^n$.

Linear non-hom:

$a_n = Ca_{n-1} + dr^n$. $\quad a_0 = \lambda$.

$a_n = Ca_{n-1}$ $\quad |C| \neq |r|$

$= C^n \lambda$

General soln:

$a_n = B(\lambda C^n) + Ar^n$ $\quad C \neq r$

$A = \frac{dr}{r-C}$

$a_n = B \cdot C^n + An \cdot r^n$ $\quad C = r$ $\quad |C| = |r|$?

$(-1)^n A + B$

Quadratic non-hom.

$C_0 a_n + C_1 a_{n-1} + C_2 a_{n-2} + dr^n = 0$.

$a_n^{(p)} = Ar^n$.

$a_n^{(h)} =$

- $A_1 r_1^n + A_2 r_2^n$, $r_1 \neq r_2 \neq r$
- $A_1 r_1^n + A_2 r_2 n$, $(r_1 = r$ or $r_2 = r)$
- $A_1 r_1^n + A_2 r_2 n$, $(r_1 = r_2 \neq r)$
- $A_1 r_1^n + A_2 r_2^n$, $(r_1 = r_2 = r)$

① → $a_n = A r^n + A_1 r_1^n + A_2 r_2^n$

② → $a_n = A_n r^n + A_1 r_1^n + A_2 r_2^n$.

③ → $a_n = A r^n + (A_1 n + B) r_1^n$

④ → $a_n = A n^2 r^n + (A_1 n + B) r_1^n$

cannot have conflict with $A_1 n + B$

homogeneous clash with particular.

① recurrence can't be satisfied & ② No extra info. from homogeneous to satisfy base cond.

so, boost up particular

reason

③ particular solution consumed by homogeneous part, the $dr^n = 0$ can't be satisfied.

when $r_1 = r$

$A r^n$

$A r_1^n$ soln of homogeneous part,

so, $0 = dr^n$

particular soln not satisfied

$Ar^n + B$ in particular soln